

AAKAAR — Roadmap (v1)

What's planned next, in roughly the order it will land. Phases are calendar guidance, not contracts; items can move when reality intervenes. Every item below is anchored to a concrete tenant pain or platform need we have already seen in v1.

Phase 1 — within 2 weeks

Goals: lower friction for new tenants, sharpen the chat UX, finish the long tail of v1 polish.

1.1 Auto-grant session capabilities on user creation

Today, a tenant admin must explicitly grant `cap.web_login` to a user before the user can run any login-bearing workflow. In practice, every user needs it. Phase 1 auto-grants the read-only "session" capabilities (`web_login` , `http.fetch_session` , `screenshot`) on user creation, with an opt-out toggle.

1.2 Vault redesign for non-login capabilities

The current vault is login-centric. Phase 1 widens the schema so a tenant can store API keys, OAuth refresh tokens, and per-site cookies — anything a capability declares it needs. The UI moves to a generic "Site → Credential type → Entry" tree.

1.3 In-chat credential widget

When the planner pauses with a `picker` signal because the vault is missing an entry, the chat will render a credential entry form inline rather than redirecting the user to `/admin/grants` . The form posts directly to the vault and resolves the signal in one step.

1.4 Multi-step login

Some sites require username on page A, password on page B, OTP on page C. The current `cap.web_login` capability assumes a single form. Phase 1 introduces `LoginFlow` as a chain of login steps a capability can declare.

1.5 Activate the capability semantic index (read path)

The FAISS index of granted capabilities is already maintained on every grant change but never read at plan-time. Phase 1 wires `CapabilityIndex.search(prompt, k)` into `PromptBuilder` so that tenants with many grants get a tighter, relevance-filtered catalog rendered into the planner prompt. For tenants with a handful of grants, behavior is unchanged.



Phase 2 — within 1 month

Goals: production hardening. Replace dev shortcuts with services that scale, add the observability story, and unlock more expressive DAGs.

2.1 OpenTelemetry observability

Phase 2 introduces full OpenTelemetry support across both backends and the frontend.

2.1.1 What gets instrumented

Layer	Spans	Notable attributes
FastAPI	one root span per request	<code>http.method</code> , <code>http.route</code> , <code>tenant.id</code> , <code>user.id</code> , <code>request_id</code>
Planner	spans for "plan" plus child spans per LLM call and tool call	<code>planner.strategy</code> (oneshot or agentic), <code>planner.candidate_count</code>
Validator	one span per DAG validation	<code>dag.node_count</code> , <code>validator.failed_rule</code>
Executor	one span per run, child spans per node, grandchild per signal wait	<code>run.id</code> , <code>node.id</code> , <code>capability.id</code> , <code>signal.kind</code>
Capability	one span per capability invocation	<code>capability.id</code> , <code>vault.read_count</code>
Browser worker	span per Playwright step	<code>browser.action</code> , <code>browser.selector_strategy</code> , <code>browser.duration_ms</code>
HTTP worker	span per outbound request	standard HTTP semantic conventions
Frontend	spans for navigation, query fetches, run-event SSE lifecycle	<code>page.path</code> , <code>query.key</code>

2.1.2 Metrics

Metric	Type	Dimensions
<code>aakar.runs.started</code>	counter	<code>tenant.id</code> , <code>capability.id</code>
<code>aakar.runs.duration_ms</code>	histogram	<code>tenant.id</code> , <code>capability.id</code> , <code>status</code>
<code>aakar.runs.failed</code>	counter	<code>tenant.id</code> , <code>capability.id</code> , <code>failed_node_id</code>
<code>aakar.signals.published</code>	counter	<code>signal.kind</code>
<code>aakar.signals.wait_ms</code>	histogram	<code>signal.kind</code>
<code>aakar.planner.tokens_in</code>	counter	<code>planner.strategy</code>
<code>aakar.planner.tokens_out</code>	counter	<code>planner.strategy</code>
<code>aakar.browser.pool.in_use</code>	gauge	(none)
<code>aakar.vault.reads</code>	counter	<code>tenant.id</code>

Histograms use the OTel default boundaries; we will tune buckets after a week of real data.

2.1.3 Logs

The structured logger (`structlog`) is bridged into OTel logs. Every log record inherits the active span context, so a log line in the executor automatically carries the `run.id` and `node.id` of the node that emitted it. Trace-to-log correlation lets ops jump from a slow span to the exact log lines for that node.

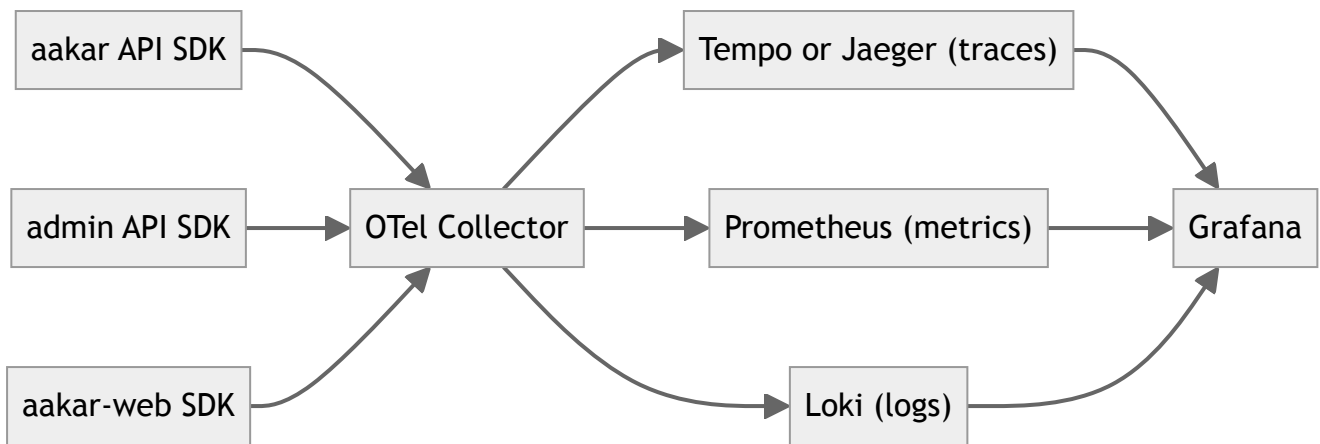
2.1.4 Resource attributes

Every signal (trace, metric, log) carries a stable resource bundle:

- `service.name` — `aakar-api` , `admin-app-api` , `aakar-web` .
- `service.version` — Git short SHA from build.
- `deployment.environment` — `dev` , `staging` , `prod` .
- `service.instance.id` — pod name in cloud, hostname in dev.

`tenant.id` is allowed as a span and metric attribute (low cardinality at our scale). `run.id` is allowed on spans only (high cardinality).

2.1.5 Pipeline and exporters



The collector runs as a sidecar in dev and as a per-node DaemonSet in cloud. Backends are pluggable; the exporters above are the v1 picks but the choice does not leak into application code.

2.1.6 Sampling

Default is head-based at 10% for normal traffic plus always-on for any trace whose root span's `error` attribute is true. Tail-based sampling on the collector retains slow traces (greater than the 95th percentile) and traces from a configurable list of "VIP" tenants for debugging.

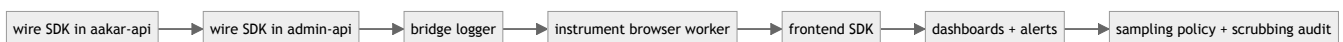
2.1.7 Privacy

Span attributes are never allowed to carry credentials, cookie values, vault payloads, or full file contents. A small allowlist plus `process.env.OTEL_SCRUB_KEYS` controls scrubbing at SDK level. PII fields in run-event payloads are redacted before they ever reach the OTEL pipeline.

2.1.8 Local developer experience

`docker compose up otel` brings up the collector, Tempo, Prometheus, Loki, and Grafana with seeded dashboards. Hitting the API in dev produces traces that show up in Grafana within seconds. The full stack is opt-in; running without `OTEL_EXPORTER_OTLP_ENDPOINT` set is a no-op.

2.1.9 Rollout sequence



2.2 Temporal-backed executor

The Executor Protocol is documented with a `TemporalExecutor` placeholder. Phase 2 finishes it, with workflow types per capability and child workflows for sub-DAGs. Local dev keeps `LocalExecutor`.

2.3 Cron triggers

Expose a per-workflow cron field. The admin surface gets a "Schedules" page. Cron is backed by the same Temporal cluster that powers the executor.

2.4 Production vault

Replace the dev plaintext-on-disk adapter with a managed-KMS-backed driver: per-tenant data key encrypts the secret blob; the data key is wrapped by a master key in a managed KMS; the master key never enters the application process. Cipher rotation is staged so old entries can be re-encrypted without downtime.

2.5 Warm browser pool

Today every run boots a fresh Chromium context (about 1.5 to 3 seconds). Phase 2 adds a pool of pre-warmed contexts isolated by tenant. Eviction on tenant-scope conflicts is enforced.

2.6 Branch and for_each in DAGs

The DAG schema gets two new node types: `branch` (route by predicate) and `for_each` (fan out over a list and join). The validator and executor learn to handle them; the planner learns to emit them when prompts call for "for each row in the file ...".



Phase 3 — 3 to 6 months

Goals: turn AKAAR from a single-deployment product into a platform that can host external publishers and serve regulated tenants.

3.1 Capability marketplace

A signed-publisher model where third-party teams can submit capabilities. Each submission goes through code review, schema validation, a security scan, and a test in a sandbox tenant before becoming visible.

3.2 Audit log surface

Today the `audit_log` table is queryable by SQL but not exposed in product. Phase 3 builds a tenant-admin audit view with filter by user, action, and time, plus signed-export for compliance.

3.3 Run replay

Given a `run_id`, replay produces a deterministic re-run from persisted DAG and inputs against a recorded mock of the third party. Useful for incident triage and regression tests.

3.4 Multi-region

The product currently assumes a single deployment per environment. Phase 3 introduces region affinity: a tenant is pinned to a region; cross-region queries are explicit; the vault is region-local.

3.5 Org hierarchy

Tenants can be grouped into orgs. Capability grants and quotas can be set at the org level and inherited; per-tenant exceptions still work.



Spike candidates (un-prioritized)

- Streaming LLM responses in chat with cancellation and partial DAG previews.
- Planner self-critique pass before emitting a DAG.
- Capability "lint" checks (selector freshness, fixture currency).
- A small mobile companion app for captcha resolution and run notifications.
- Feature-flag service plus per-tenant rollout percentages.
- Out-of-the-box SAML / OIDC.

Engineering policies

- **One PR per logical change.** Bundle only when a split would be churn.
- **Integration tests hit a real database.** No mocks at the persistence boundary.
- **Migrations are reviewed for online-safety.** Adding a NOT NULL column to a hot table is a multi-PR sequence.
- **No silent fallbacks for security.** A missing capability or a missing grant returns an explicit error.
- **Comments justify the WHY only.** WHAT is in the code; PR description carries context.
- **No hard-coded credentials anywhere.** Vault or env, never in code or tests.

Decision log (open questions)

ID	Question	Owner	Target decision date
RFC-04	Are we ok with shipping <code>TemporalExecutor</code> before warm-pool isolation is proven?	Platform	2026-05-30
RFC-05	Where does run-replay state live: same DB or separate journaling store?	Platform	2026-06-15
RFC-06	Cron triggers in admin-app or in aakar API?	Platform	2026-05-25
RFC-07	OTel span naming: do we adopt OpenTelemetry Semantic Conventions verbatim or define an AKAAR overlay?	Observability	2026-05-20

What this roadmap is not

- It does not promise dates. Items move when scope changes or facts on the ground move.
- It does not authorize work outside the listed phases without an explicit RFC.
- It does not replace the HLD or LLD. Items here that touch the architectural spine require a corresponding HLD or LLD update before merge.